Kritazya Upreti

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Building full-stack products and AI systems, from LLM-powered apps down to embedded firmware

EDUCATION

University of Texas at Arlington Arlington, TX

B.S. in Computer Science, Minor in Mathematics | GPA: **3.90/4.00** Expected Dec 2027

- **Coursework:** Data Structures and Algorithms, Operating Systems, Databases, Artificial Intelligence, Computer Networks
- **AI Engineering (CodePath)**: 10-week program, completed with Honors distinction

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, C#, C, SQL

AI/ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, CNNs, RAG, LangChain, LangGraph, LLM APIs, ChromaDB, Pinecone

Backend & Full-Stack: FastAPI, Node.js, Express, React, REST APIs, PostgreSQL, JWT/OAuth, Chrome MV3, Vite

Developer Tools & Infrastructure: Linux (Ubuntu), Git, GitHub Actions (CI/CD), Docker, Claude, Codex, Render, Vitest, Jasmine

Robotics & Embedded: ROS 2, ArduPilot, Gazebo/SITL, ESP32, Arduino, UART

EXPERIENCE

Research Assistant | Python, NumPy, Pandas, Matplotlib, C++, C#, ESP32, Machine Learning Aug 2024 – Aug 2026

University of Texas at Arlington Research Institute Fort Worth, TX

- Built an end-to-end Python ML pipeline for soft-robotic pressure prediction, collecting and preprocessing sensor data and benchmarking 10 regression models; achieved **99.55% test R^2** and **92.73% cross-validation R^2**
- Wrote embedded C++ firmware for ESP32 microcontrollers to acquire sensor data and drive valve and actuator control over serial (UART), enabling real-time control of a soft robotic hand
- Built a VR rehab game synced to a soft robotic glove and maintained a Unity/C# 3D rehab game synced to physical controllers and sensors; fixed its integration issues for **zero-failure live demos** to 30+ students and industry representatives
- Co-authored a peer-reviewed paper on a learning-based soft pneumatic wrist exoskeleton, published at **AIRC 2025**

ROS 2 Team Lead - Autonomous UAV | C++, Python, ROS 2 Jazzy, ArduPilot, Gazebo, Ubuntu, Git Aug 2026 – Present

University of Texas at Arlington

- Lead ROS 2 development for the search-and-rescue UAV subsystem, coordinating a **13-person** team through sprint planning, code reviews, and a GitFlow branching workflow
- Develop and validate autonomous search-pattern software in Python against Gazebo and ArduPilot SITL, integrating the autonomy stack with the flight controller over UDP

PROJECTS

Buddy – Agentic AI Productivity Mentor | Python, FastAPI, scikit-learn, PostgreSQL, Gemini API, ElevenLabs Sep 2026

- Built a Chrome MV3 extension and async FastAPI backend that streams browsing sessions into TigerData (PostgreSQL) via idempotent upserts, eliminating double-counted time under **40 concurrent, out-of-order heartbeats**
- Trained a TF-IDF + logistic-regression classifier on **1,129 labeled sessions** with leak-proof, template-grouped validation splits, reaching **92.5% accuracy** and **100% recall** on productive work
- Built a voice-driven LLM agent (Gemini, ElevenLabs) with tool use and persistent context, adding tiered fallbacks and confirmation-gated writes to prevent silent failures and state corruption

UTAFindR – Hybrid Retrieval | Python, BM25, sentence-transformers, cross-encoders Aug 2026 – Sep 2026

- Built a hybrid RAG pipeline that answers UTA CS students' enrollment questions, combining BM25 and dense retrieval through reciprocal rank fusion with cross-encoder reranking
- Designed fusion rules that protect literal matches, preserving **100% exact-match accuracy** on keyword-sensitive queries (course codes, policy terms), verified by an automated evaluation harness

Momentum – Full-Stack Habit-Tracking Platform | React, Node.js, Express, PostgreSQL, Vitest Jun 2026 – Aug 2026

- Shipped a production web app on Render with **30 REST endpoints** over a normalized 7-table PostgreSQL schema, using JWT authentication with HttpOnly cookies, bcrypt hashing, and GitHub OAuth.
- Wrote a **47-test Vitest suite** covering auth and core flows; diagnosed and hotfixed a production bug that prevented affected users from completing their first week of onboarding.
- Built a rule-based engine generating personalized daily workouts from a 96-exercise library, saved in one atomic transaction.

DocuExtract – Mortgage Document AI | FastAPI, React, TypeScript, ChromaDB, Tesseract Oct 2025

- Co-built a FastAPI + React/TypeScript app that OCRs uploaded mortgage documents, extracts key-value fields from single-page files, and routes multi-page files into a RAG index for Q&A
- Indexed **14,383 chunks** from 500+ pages of Fannie Mae, Freddie Mac, FHA, and USDA guidelines in ChromaDB with retrieval latency under 500 ms, falling back to live web search only when no document matches.

HONORS

$5,000 GenSpiration Prize, National Academy of Inventors – Won $5,000 for a 2-degree-of-freedom soft robotic wrist exoskeleton invention, including the control algorithm, working prototype, and GUI.